

WASP-39b

R-1731

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_130521 · created 2026-08-02T17:36:34+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456433.833 +/- 0.038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.049
Transit depth (Rp/Rs)^2	2.18 +/- 1.44 [%]
Orbital Inclination (inc)	88.26 +/- 2.12
Airmass coefficient 1 (a1)	0.982 +/- 0.016
Airmass coefficient 2 (a2)	0.0092 +/- 0.0084
Scatter in the residuals of the lightcurve fit is	1.61 %
Best Comparison Star	#2 - [401, 45]
Optimal Method	PSF photometry
Transit Duration (day)	0.09 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.148 ± 0.049 vs 0.1457 (deviation 0.05σ)
Duration fit vs published	0.09 d vs 0.12 d (deviation 0.91σ)
Mid-time O–C	-10.0 min
Depth SNR	3.0
Residual scatter	1.61 %

Light curve

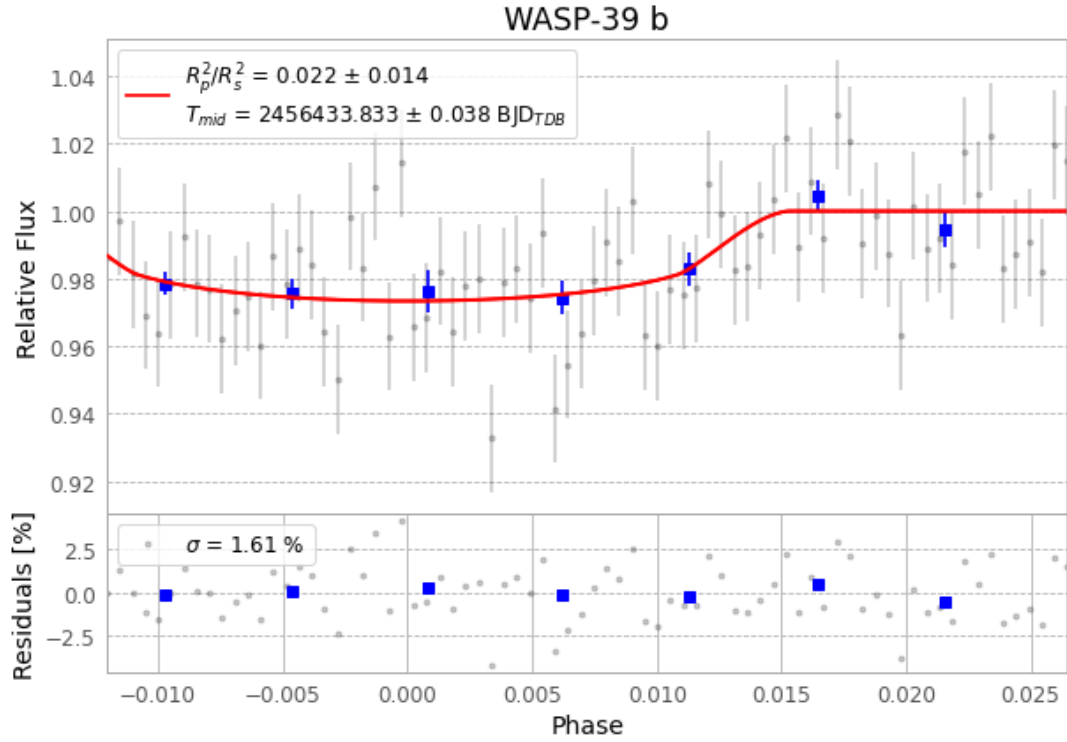


Figure 1 — FinalLightCurve_WASP-39 b_2013-05-20.png (SHA-256 1e21ca22f58c3fe2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [313, 291], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 312ce659396602ff...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-39130521063912.FITS → WASP-39130521102412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-130521.log (b0e69d47a4fa...), FinalLightCurve_WASP-39 b_2013-05-20.png (1e21ca22f58c...), FinalLightCurve_WASP-39 b_2013-05-20.csv (590ee53e3129...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 17:36Z.